README

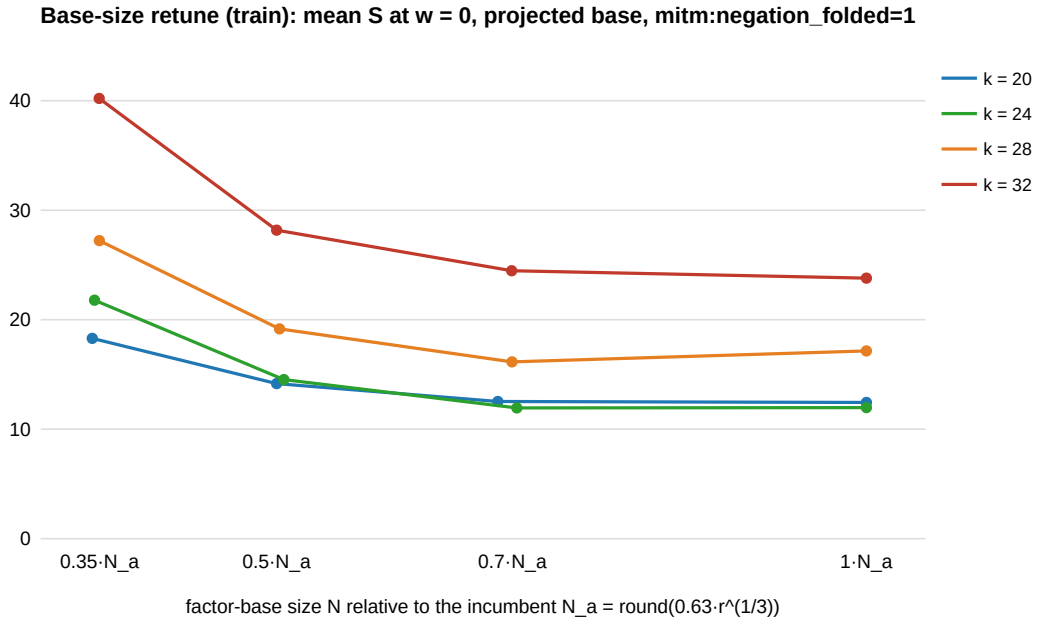
Curve25519-family IC ratchet round 1: four leads (2026-11-02)

The incumbent is the projected small-x base with `mitm:negation_folded=1` (see `../curve25519_ic_baseline_20261102`). Protocol: PROTOCOL.md, frozen before data. PDF: REPORT.pdf.

Outcome: no lead improves the baseline, so the incumbent stays. Each lead was measured or settled exactly, and each null has a named reason.

lead	track	result	class
1. Retune N	generic, ecbench end-to-end S	train picked N_a at k = 20, 32 and 0.7·N_a at k = 24, 28. Holdout sel/inc = 0.996 [0.944, 1.051] (k = 24) and 0.977 [0.941, 1.015] (k = 28) at w = 0, reversing to 1.21 at w = 1	**null** (band unresolved); incumbent kept
2. Closed cofactor fiber	generic, exact counts	covers exactly the projected base's targets (D equal at all 4 sizes) with **8.0×** the table entries	**dominated** by the incumbent (F1, F2 hold); no ecbench arm
3. Base closed under x ↦ 1/x	generic, exact counts (T_0 study)	2.0× over the blind base, 4× below projection; on a projected base, closure is a no-op because `[8](P+T₀) = [8]P`	**dominated**; useful only on the algebraic track
4. Algebraic (summation polynomial + F4)	algebraic, ecbench smoke	`pkm-semaev` needs `2^t | p − 1`; Curve25519 primes have `p ≡ 5 (mod 8)`, so t ≤ 2 and the base has 3 columns at k = 20. One verified solve: S = 1027 (m = 2) / 211 (m = 3) at w = 0, and $4.9 \cdot 10^4$ / $6.2 \cdot 10^6$ at w = 0.1, against 9.05 / 9.46 for the incumbent	**blocked**: problem record

1. Base-size retune



sweep

Train (16 targets, 1 round, fresh seeds). Mean S at w = 0 for N = 0.35 / 0.5 / 0.7 / 1.0 · N_a:

k	N grid	mean S
20	11 / 16 / 22 / 32	18.30 / 14.16 / 12.53 / **12.44**
24	28 / 41 / 57 / 81	21.79 / 14.53 / **11.94** / 11.98
28	71 / 102 / 142 / 203	27.22 / 19.17 / **16.15** / 17.14
32	179 / 256 / 358 / 512	40.21 / 28.18 / 24.48 / **23.79**

Holdout (32 fresh targets × 2 rounds; inc = N_a, sel = N*):

k	N_a → N*	S inc → sel (w = 0 / 0.1 / 1)	sel/inc, paired [95%] at w = 0	band verdict
24	81 → 57	12.52 → 12.47 / 13.64 → 14.08 / 23.68 → 28.58	0.996 [0.944, 1.051]	unresolved
28	203 → 142	16.60 → 16.22 / 18.21 → 18.54 / 32.74 → 39.46	0.977 [0.941, 1.015]	unresolved

At k = 20 and 32 the train step selected N_a itself, and the holdouts record the A/A null. The guess that the borrowed size was too large is not supported. S is flat from 0.7·N_a to N_a and rises steeply below. A smaller N trades table work for more probes, whose unpriced lookups lose at w = 1.

2. Closed cofactor fiber (`curve25519_family fiber`)

k	N	D blind	D closed fiber	D projected	table entries fiber / projected
20	32	154	1003	1003	8192 / 1024 = 8.0
24	81	851	6552	6552	52488 / 6561 = 8.0
28	203	5101	41158	41158	329672 / 41209 = 8.0
32	512	32755	262030	262030	2097152 / 262144 = 8.0

`S ↦ S − t(S)` (removing the sum's own torsion component) and `S ↦ [8]S` put the same targets in reach, so the two counts are equal by construction. The difference is cost: the fiber table has 8× the entries. The "about 2×" quoted earlier for this lead was against the blind base. Against the incumbent it loses.

4. Why the algebraic track is blocked on Curve25519

PKM's multiplicative base `x ∈ g·μ_n` needs a large, smooth `n | p − 1` that is a tower of low-degree maps. For `p = 2^255 − 19`, `p − 1 = 2²·3·65147·q` (q a 236-bit prime), so the only such tower has order 12. For the miniatures, `p ≡ 5 (mod 8)` by construction, so `2^t | p − 1` stops at t = 2. Curve25519's choice of prime closes PKM's Kummer route. Only the isogeny form remains, and the lab has its base but no Semaev oracle for it. The T_0-closed algebraic base would want `μ_{2^t}` in Montgomery coordinates (closed under inversion), which is the object this same congruence forbids. The unblock conditions are in the problem record.

Distance to rho (unchanged)

The incumbent's distance to matched rho stays 7.8× / 10.6× / 18.1× / 24.7× (baseline study). The holdout's own rho arms agree: k = 24 inc ÷ rho = 10.5×; k = 28 15.1×.

Evidence

All sessions verified. Audits: `analysis/verify-{train,holdout}-k*.json` (8 and 12 replays) and `verify-pkm-smoke-k20.json`, all OK. Comparisons: `analysis/compare-holdout-k{24,28}.txt` and `sessions/holdout-k*/comparisons/`. Exact counts: `analysis/fiber.jsonl`. Records ≥ 64 KiB are in the artifact store (`.artifacts.json`). The route deviation is as in the baseline study: local host, counted units, no wall-time claim.

Next

- The generic track is saturated at m = 2 on this family. Known remaining levers apply to any prime curve, not to Curve25519's structure: m = 3 (`mitm:m=3` and `rootfind-mitm:m=3`), and large primes on the pair table.
- The algebraic track needs the isogeny-tower oracle (problem record).

PROTOCOL

Curve25519-family IC ratchet round 1 — protocol

Frozen 2026-11-02 before any data in this directory. The incumbent is the baseline of ../curve25519_ic_baseline_20261102: `prime-projected-abscissa:size=N_a,scan_cap=16N_a` + `mitm:negation_folded=1`, with `N_a = 32, 81, 203, 512` at k = 20, 24, 28, 32. Curves are the same miniatures. Every target seed below is new.

The owner asked for four leads (2026-11-02): (1) retune N, (2) the closed cofactor fiber for prime curves, (3) a base closed under x ↦ 1/x, and (4) the algebraic (summation-polynomial) track. This protocol covers all four. Each part is preregistered before its data is taken.

Part 1 — base-size retune (generic track, ecbench, end-to-end S)

- **Train.** One session per k with the projected base at `N ∈ round({0.35, 0.5, 0.7, 1.0}·N_a)`, plus rho-neg: 16 targets × 1 round, target seed `2026110400 + k`. `N*` = the lowest mean S at w = 0.
- **Holdout.** One session per k with `inc` (N_a), `sel` (N*), `inc-aa`, `rho-neg` and `rho-neg-aa`: 32 targets × 2 rounds, target seed `2026110500 + k`. If `N* = N_a`, the holdout still runs and records a null.
- **Decision.** `sel` is an improvement at size k if the paired `ecbench compare` ratio sel/inc has its 95% interval below 1 at w = 0 **and** `ecbench band` ranks sel < inc across w ∈ {0, 0.1, 1}. Otherwise the result is relabelling (better at w = 0 only) or null. Prediction: small gains (≤ 15%) with N* ∈ {0.5, 0.7}·N_a, since the incumbent's table and relation phases are already about balanced (49% / 47% of GAE at k = 32).

Part 2 — closed cofactor fiber on prime curves (mechanism, exact counts)

The binary-curve `closed` fiber mode stores `P_i ± P_j + K` for all `K ∈ E[8]`, so one probe covers all 8 cosets. On the blind small-x base this recovers the full cofactor. **Stated before measuring:** it covers the same targets per column as the projected base (`S ↦ [8]S` is a bijection from coset representatives to G), but its table is 8× larger. So against the incumbent it is predicted to lose: equal yield, 8× table build. Measured with `curve25519_family fiber` (exact `D` for the blind, closed-fiber and projected tables at N = N_a, and table entries for each).

- **F1.** `D_fiber / D_proj` ∈ [0.95, 1.05].
- **F2.** `entries_fiber / entries_proj` ∈ [7.5, 8.5].

If both hold, the closed fiber is dominated by projection in the generic track and gets no ecbench arm.

Part 3 — base closed under x ↦ 1/x

Already measured for the generic track in ../curve25519_t0_symmetry_20261102: 2.0× over blind, 4× below projection. In the generic track a projected base gains nothing from closure, because `[8](P + T₀) = [8]P`. Part 3 therefore runs inside Part 4, where a small-x base is required and cannot be projected.

Part 4 — algebraic track

The lab's prime-field algebraic oracle is `pkm-semaev` on a `prime-kummer:t=T` base (Petit–Kosters–Messeng tower, S_(m+1) solved by F4). Its arms and sizes are fixed in an amendment written after a feasibility check: one smoke target, with no timing used for any decision. The comparison is pkm-semaev against the generic incumbent at equal k, end-to-end S, and the §5a band, on fresh targets `2026110600 + k`. The T_0-closed variant is in scope only if the pkm-semaev base admits closure under x ↦ 1/x without changing the solver; if not, it is named as an obstruction with what would unblock it.

PROBLEM

Curve25519-family algebraic index calculus: no usable PKM tower

- **Status:** `open`
- **Kind:** proved obstruction (multiplicative Kummer tower) and implementation gap (isogeny-tower Semaev oracle)
- **Affected work:** ratchet round 1, Part 4; the T_0-closed algebraic base proposed in `curve25519_t0_symmetry_20261102`
- **Next action owner:** unassigned
- **Last updated:** 2026-11-02

Problem

We cannot yet measure an algebraic-track (summation polynomial plus factor-base constraint, solved by F4) index calculus on Curve25519-structured curves at a useful base size. The lab's only prime-field algebraic oracle, `pkm-semaev`, requires the Kummer base `x ∈ g·μ_{2^t}`, which needs `2^t | p − 1`. Every Curve25519-family prime has `p ≡ 5 (mod 8)`, so $v_2(p-1) = 2$ and $t \le 2$: at most 4 abscissae.

Evidence and uncertainty

- **Observed:** `research/curve25519_ic_ratchet_20261102/sessions/pkm-smoke-k20` (k = 20 miniature, one target, `ecbench verify --replay 4` OK). With `prime-kummer:t=2` the base has 3 columns; `pkm-semaev:m=2,mode=full` costs S = 1027 / 49,513 / 485,893 and `m=3,mode=chain` costs S = 211 / $6.18\cdot10^6$ / $6.18\cdot10^7$ at w = 0 / 0.1 / 1. The generic incumbent costs S = 9.05 / 9.46 / 13.16 on the same target.
- **Derived or proved:** `p ≡ 5 (mod 8)` ⇒ $v_2(p-1) = 2$. For Curve25519, $p - 1 = 2^2\cdot3\cdot65147\cdot q$ with `q` a 236-bit prime (GNU `factor`). So the largest subgroup of `F_p^*` reachable as a composition of power maps of degree ≤ 3 has order 12. Multiplicative PKM bases (any smooth order) are limited to this, plus one prime-degree step of 65147.
- **Suspected explanation:** not a suspicion. The structure of `p − 1` decides it.
- **Missing capability:** an oracle that writes `S_{m+1}` against PKM's *isogeny-form* tower (`prime-isogeny:t=T` exists as a base, with zeros of a chain of Vélu 2-isogeny x-maps, and needs no condition on p). `f4_fp_tower` supports only the squaring rule $y_j^2 = y_{j+1}$.
- **What this evidence does not establish:** it says nothing about whether an isogeny-tower algebraic oracle would beat the generic incumbent. In the lab's prime-order studies the algebraic track trails the generic track, so the prior is that it would not.

What would unblock it

- [ ] `pkm-semaev` (or a sibling) accepts `prime-isogeny:t=T`, with the tower ring relations `y_{j+1} = φ_j(y_j)` for the degree-2 Vélu x-maps, its refutations cross-checked on the grid (`crosscheck=1`), and unit tests on a curve with a known decomposition.
- [ ] The same, run on the k = 20 and k = 24 miniatures at `t` with `2^t ≈ N_a`, every solve verified, `ecbench verify --replay` OK.

Meeting both makes the comparison possible; it does not predict its outcome. The T_0-closed variant needs a tower whose zero set is closed under `x_M ↦ 1/x_M`. `μ_{2^t}` in Montgomery coordinates is closed under inversion, but it is exactly the object `p ≡ 5 (mod 8)` forbids. An isogeny tower would need to be chosen compatibly with the involution, which is open.

Next discriminating step

- **Action:** implement the isogeny-form tower ring in `f4_fp_tower` (Vélu x-maps as rewrite rules) behind `pkm-semaev`, with the grid cross-check.
- **Inputs and controls:** the k = 20 miniature; a prime-order control curve where both the Kummer and isogeny forms exist (`2^t | p − 1`), so the two towers can be compared at equal `t`.
- **Execution limits and stop condition:** `p < 2^31` (engine limit); timeouts recorded as such.
- **Outcomes and next actions:** if it costs within 10× of the incumbent across the band, run a frozen holdout. If not, record the gap as the algebraic track's distance remaining and stop.
- **Recheck trigger:** any change to `f4_fp_tower` adding non-squaring tower rules.

Resolution and history

- 2026-11-02: opened from ratchet round 1, Part 4. Unresolved.